

Evaluation report on Voice Assistant

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1 Requirements Analysis

1.1 Functional Requirements

1. System must be able to process English voice as input and produces spoken English as output.
2. Users can perform CRUD operations on calendar API through speech.
3. The system should keep track of conversation history.

1.2 Non-Functional Requirements

1. The system must be able to perform CRUD operations on calendar API correctly as defined by the API specification.
2. Evaluation on accuracy and reliability.
3. Support english language.

1.3 Constraints

1. Voice assistant must run locally only external connections weather and calendar APIs are allowed.
2. Final product must be delivered as Docker container.
3. The project must be completed in a group of 4-5 students.

2 Evaluation

2.1 WER(Word Error Rate) Accuracy

In the evaluation, we measured the performance using Word Error Rate to quantify the differences between the ground truth and system's hypothesis. We manually tested the sentences in Table 1 and calculated the WER for each of them.

$$WER = \frac{S + D + I}{N} \quad (1)$$

2.2 Slot-Filling Accuracy(The Understanding)

In this section, we evaluated if the system parses the transcription successfully. For each sentence we check if the slots location, date, intent were correctly identified.

$$SER = \frac{\text{Total Incorrect Slots}}{\text{Total Expected Slots}} \quad (2)$$

2.3 Task Success

This evaluation looks at the result. Did the user get what they want?

For example: Verify if the system actually adds an appointment or returns the weather forecast correctly.

Ground Truth (Reference)	ASR Transcription (Hypothesis)	Extracted Slots	WER	Task Success
What is the weather forecast for Berlin tomorrow?	What is the weather forecast for Berlin tomorrow?	{'day': 1, 'location': 'Berlin'}	0	Pass
I need the weather description for Tokyo for the day of Tuesday.	I need the weather description for Tokyo for the day of Tuesday. Response: The weather in Tokyo on tuesday is shower rain, with temperatures between 17 and 20 degrees Celsius.	{'day': 2, 'location': 'Tokyo'}	0	Pass
What is the highest temperature expected for Paris on Saturday?	What is the highest temperature expected for Paris on Saturday?	{'day': 6, 'location': 'Paris'}	0	Pass
Add a Doctor Visit on my calendar for the twelfth of January.	Add a doctor visit on my calendar for the 12th of January.	{'action': 'create', 'day': 345}	0	Pass
Find my next appointment and tell me the title of it.	find my next appointment and tell me the title of it.	{'action': 'list', 'list-mode': 'next', 'reference': 'previous'}	0	Pass
Move my appointment Lunch with Sarah from Monday to Tuesday.	Move my appointment lunch with Sarah from Monday to Tuesday.	{'action': 'update', 'day': 1, 'title': 'lunch with Sarah from Monday to Tuesday', 'location': 'Tuesday'}	0	Fail

Table 1: System Evaluation (WER, Understanding, and Task Success)

2.4 Summary

This initial evaluation gives 0% Word Error Rate. This task-oriented ASR system uses rules to complete tasks, that's why we used only task-related sentences for testing and it's working well only on those tasks.